

Considering the difficulties and challenges with mobile strategies, enterprises are looking for a unique and cost-effective solution that reduces the time to market and offers high maintainability. Rapid mobile application development is one of the basic needs from business heads, without limiting the features or quality of the deliverables. Rather, mobile solutions are assumed to meet dynamic mobile environment challenges, integration with other systems/frameworks/modules with expected add-ons like an EMM suite, push notifications, SSO (single sign on) for mobile solutions, etc. Considering the dynamic nature of system needs, mobile hybrid applications are trending across the market for the B2C, B2E and B2B segments.

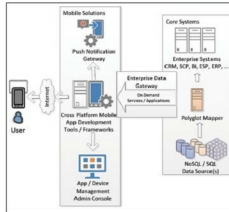


Figure 3: Solution overview

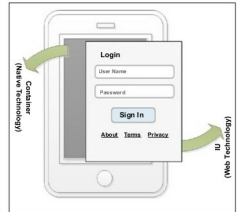


Figure 4: Mobile hybrid application

Mobile hybrid applications

Mobile hybrid applications are seamless amalgamations of native (iOS, Android or Windows Phone) and Web (HTML, CSS and JavaScript) technologies to provide multi-platform supported mobile applications in a cost and time effective manner. Hybrid applications are neither native apps nor Web apps. They are implemented with Web technologies and packaged as applications for distribution. These apps can access native device features and APIs. Basically, hybrid apps are native mobile applications which have Web browser control within their main UI screen. Here are the imperative characteristics for mobile hybrid apps:

- UI implementation using Web technologies (HTML, CSS and Java Script)
- Apps that are capable of using mobile platforms or operating system APIs
- An executable file that installs and resides in the mobile device
- Apps that can be distributed via platform-specific app stores or enterprise distribution mechanisms

Figure 4 depicts an analogy of a hybrid application.

Broadly, it has two layers—the *container* and the *user interface*. The *container* is implemented on the native mobile technology (Android, iOS or Windows Phone), whereas the *UI (User Interface)* is implemented with Web technologies (HTML, CSS and JS).

There are many frameworks and tools for hybrid app development. The major selling point for hybrid apps is cross-platform development. The following are the features that can be leveraged with hybrid apps:

- Integration of open source frameworks with HTML5
- Liquid layouts for multi-screen UIs
- Local storage, multimedia handling, semantics and forms,

graphics, etc

- A single code-based architecture model for multi-platform presentations
- Hybrid framework that bundles the HTML5 based view layer with native platform containers to create deployable builds

Use case

In this era of globalisation, there are many factors other than technology, hardware or software, which directly or indirectly influence solution approaches. The key for next generation mobile applications is to be available *anytime and anywhere*. Enterprise 'search on mobile' is one of the use cases for this. Consider a scenario where the user's structured and unstructured data can reside in mobile devices. It is accessible online as well as offline. There is automatic data fetch and synchronisation from enterprise servers, while hybrid mobile applications support security mechanisms with adequate security enforcement for enterprise users.

Enterprise users (CEOs, CIOs, CTOs, executives, sales representatives, employees, etc) can thus have their structured or unstructured data (presentations, documents, contact details, agreements, statements of work, etc) on a mobile device, anytime and anywhere, either online or offline. **END** 

References

- [1] 'Kundera', Github <<https://github.com/impetus-opensource/Kundera>> accessed 09 February 2015.
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